Math 1232 Summer 2025
Mastery Quiz 5
Due Thursday, July 17

This mastery quiz has three topics. Everyone should do S4, but if you already have a 4/4 on M2 or a 2/2 on S3, feel free to skip these topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in class on Thursday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 2: Integration Techniques
- Secondary Topic 3: Improper Integrals
- Secondary Topic 4: Arc Length and Surface Area

Name: Solutions

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Major Topic 2: Integration Techniques

(a) Compute $\displaystyle\int e^x \sin(5x)\,dx$.

Solution.

$$\begin{aligned}
\int e^x \sin(5x)\,dx &= e^x \sin(5x) - \int 5e^x \cos(5x)\,dx \\
\int e^x \cos(5x) &= e^x \cos(5x) + \int 5e^x \sin(5x)\,dx \\
\int e^x \sin(5x)\,dx &= e^x \sin(5x) - 5e^x \cos(5x) - 25\int e^x \sin(5x)\,dx \\
\implies 26\int e^x \sin(5x)\,dx &= e^x \sin(5x) - 5e^x \cos(5x) + C \\
\int e^x \sin(5x)\,dx &= \frac{1}{26}e^x \sin(5x) - \frac{5}{26}e^x \cos(5x) + C.
\end{aligned}$$

(b) Compute $\displaystyle\int \frac{3x}{(x+4)(x-2)}\,dx$.

Solution.

$$\begin{aligned}
\frac{3x}{(x+4)(x-2)} &= \frac{A}{x+4} + \frac{B}{x-2} \\
3x &= A(x-2) + B(x+4) \\
6 &= 6B \Rightarrow B = 1 \\
-12 &= -6A \Rightarrow A = 2 \\
\frac{3x}{(x+4)(x-2)} &= \frac{2}{x+4} + \frac{1}{x-2} \\
\int \frac{3x}{(x+4)(x-2)}\,dx &= \int \frac{2}{x+4} + \frac{1}{x-2}\,dx \\
&= 2\ln|x+4| + \ln|x-2| + C.
\end{aligned}$$

Name: Solutions

(c) Compute $\displaystyle\int \frac{\sqrt{1+9x^2}}{x^2}\,dx$.

Solution. We see something almost of the form $\sqrt{a^2+x^2}$, factoring out the 9 we have $\sqrt{1+9x^2} = 3\sqrt{\frac{1}{9}+x^2}$, so we make the substitution $x = \frac{1}{3}\tan\theta$ so that $\sqrt{1+9x^2} = \sqrt{1+\tan^2(\theta)} = \sec(\theta)$. Then $dx = \frac{1}{3}\sec^2(\theta)\,d\theta$ so that

$$\begin{aligned}
\int \frac{\sqrt{1+9x^2}}{x^2}\,dx &= \int \frac{\sec(\theta)}{\frac{1}{9}\tan^2(\theta)} \cdot \frac{1}{3}\sec^2(\theta)\,d\theta \\
&= 3\int \frac{\sec^3(\theta)}{\tan^2(\theta)}\,d\theta = 3\int \frac{\frac{1}{\cos^3(\theta)}}{\frac{\sin^2(\theta)}{\cos^2(\theta)}}\,d\theta \\
&= 3\int \frac{\cos^2(\theta)}{\sin^2(\theta)\cos^3(\theta)}\,d\theta = 3\int \csc^2(\theta)\sec(\theta)\,d\theta.
\end{aligned}$$

This is a product of two functions, and we know that $\csc^2(\theta)$ is the derivative of $-\cot(\theta)$, so we do integration by parts with $u = \sec(\theta)$, $dv = \csc^2(\theta)$. Then $du = \tan(\theta)\sec(\theta)$ and $v = -\cot(\theta)$, so the integration by parts formula gives

$$\begin{aligned}
3\int \csc^2(\theta)\sec(\theta)\,d\theta &= 3\left(-\cot(\theta)\sec(\theta) - \int(-\cot(\theta))\tan(\theta)\sec(\theta)\,d\theta\right) \\
&= -3\cdot\frac{\cos(\theta)}{\sin(\theta)}\cdot\frac{1}{\cos(\theta)} + 3\int \sec(\theta)\,d\theta \\
&= -3\csc(\theta) + 3\ln|\sec(\theta)+\tan(\theta)| + C.
\end{aligned}$$

We then draw a triangle with opposite side $3x$ and adjacent side 1 so that $\tan(\theta) = 3x$. Solving for the hypotenuse with the Pythagorean Theorem gives $\sqrt{1+9x^2}$, and this tells use that

$$\csc(\theta) = \frac{\sqrt{1+9x^2}}{3x} \quad \text{and} \quad \sec(\theta) = \sqrt{1+9x^2},$$

being hypotenuse/adjacent and hypotenuse/opposite, respectively. Therefore

$$\begin{aligned}
\int \frac{\sqrt{1+9x^2}}{x^2}\,dx &= -3\csc(\theta) + 3\ln|\sec(\theta)+\tan(\theta)| + C \\
&= -\frac{\sqrt{1+9x^2}}{x} + 3\ln|\sqrt{1+9x^2}+3x| + C.
\end{aligned}$$

Name: Solutions

Secondary Topic 3: Improper Integrals

(a) Compute $\displaystyle\int_0^{\pi/2} \frac{\sin(t)}{\sqrt{\cos(t)}}\,dt$.

Solution. We know that $\frac{\sin(t)}{\sqrt{\cos(t)}}$ is undefined at $\pi/2$. So we have

$$\begin{aligned}\int_0^{\pi/2} \frac{\sin(t)}{\sqrt{\cos(t)}}\,dt &= \lim_{t\to\pi/2}\int_0^{t} \frac{\sin(t)}{\sqrt{\cos(t)}}\,dt \\ &= \lim_{t\to\pi/2} -2\sqrt{\cos(t)}\Big|_0^t \\ &= \lim_{t\to\pi/2} -2\sqrt{\cos(t)}+2 = 2\end{aligned}$$

where we have used the substitution $u = \cos(t)$, $du = -\sin(t)\,dt$.

(b) Compute $\displaystyle\int_1^{\infty} \frac{\ln(x)}{x}\,dx$.

Solution. We have

$$\begin{aligned}\int_1^{\infty} \frac{\ln(x)}{x}\,dx &= \lim_{t\to\infty}\int_1^{t} \frac{\ln(x)}{x}\,dx \\ &= \lim_{t\to\infty}\frac{1}{2}\ln(x)^2\Big|_1^t \\ &= \lim_{t\to\infty}\frac{1}{2}\ln(t)^2 - 0 = \infty.\end{aligned}$$

Thus the integral diverges. (Here we've used the substitution $u = \ln(x)$ with $du = \frac{1}{x}\,dx$.) We could also have argued as follows: for $x \geq e$, $\ln(x) \geq 1$, and therefore $\ln(x)/x \geq 1/x$. This means $\int_e^\infty \ln(x)/x\,dx$ must diverge by the comparison test since we know that $\int_e^\infty 1/x\,dx$ diverges. Hence the original integral diverges, because the interval $[1,\infty)$ we integrate on contains the interval $[e,\infty)$.

Name: Solutions

Secondary Topic 4: Arc Length and Surface Area

(a) Set up (but don't evaluate!) an integral that gives the arc length of the curve $\arctan(x) = y^3$ as x varies from 1 to 6.

Solution. We have $y = \sqrt[3]{\arctan(x)}$ so $y' = \frac{1}{3}(\arctan(x))^{-2/3}\frac{1}{1+x^2}$, and thus

$$L = \int_1^6 \sqrt{1 + \frac{1}{9\arctan(x)^{4/3}(1+x^2)^2}}\,dx.$$

(b) Compute the area of the surface obtained by taking the curve $y = \sqrt{15-x}$ as x goes from 3 to 5 and rotating it around the x-axis.

Solution. We have $y' = \frac{-1}{2\sqrt{15-x}}$. So we get

$$\begin{aligned}
A &= \int_3^5 2\pi y\sqrt{1+(y')^2}\,dx \\
&= \int_3^5 2\pi\sqrt{15-x}\sqrt{1+\frac{1}{4(15-x)}}\,dx \\
&= 2\pi\int_3^5 \sqrt{15-x+\frac{1}{4}}\,dx \\
&= \pi\int_3^5 \sqrt{61-4x}\,dx \\
&= \pi\frac{2}{3\cdot(-4)}(61-4x)^{3/2}\Big|_3^5 = \frac{-\pi}{6}\left(41^{3/2}-49^{3/2}\right) \\
&= \frac{\pi}{6}(343-41\sqrt{41}) \approx 42.13.
\end{aligned}$$